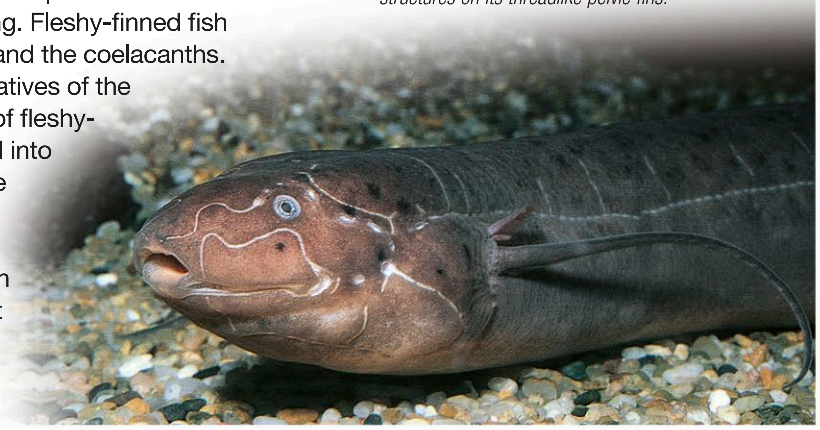
Fleshy-finned fishes

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Sarcopterygii
ORDERS	2
FAMILIES	4
SPECIES	48

Fleshy-finned fishes have enlarged fleshy fin bases that consist of muscles connected to an internal skeleton. Some species use these fins to "walk" on the seabed as well as for swimming. Fleshy-finned fish are divided into 2 groups (orders), the lungfish and the coelacanths. Lungfish are thought to be the closest living relatives of the 4-legged animals. Somewhere in the ancestry of fleshy-finned fish and tetrapods, fleshy fins developed into limbs capable of movement on land. The eellike lungfish have one or two primitive lungs as well as gills and so can breathe air if necessary They can live in oxygen-poor swamps and even survive out of water. One species, the West African lungfish, can reside in a mud cocoon at the bottom of a dry lake, awaiting the next wet season to replenish the water. The other group, the coelacanths, were thought to be extinct since the time of the dinosaurs until discovered in 1938; they live in deep areas of the Indian Ocean.

AIR-BREATHING FISHES
Larval South American lungfish have gills, while the adults have 2 lungs and can only breathe air. When in its breeding burrow, the male provides extra oxygen for its young by releasing it into the water from feathery structures on its threadlike pelvic fins.



Latimeria chalumnae

Coelacanth

Location W. Indian Ocean

Length 5–6 ft (1.5–1.8 m)
Weight 145–220 lb (65–98 kg)
Sex Male/Female
Status Critically endangered

First seen by scientists in 1938, the coelacanth is a classic example of a "living fossil," because it belongs to a group that was thought to have died out over 65 million years ago. It is found at depths of 500–2,300 ft (150–700 m) along rocky slopes with submarine caverns, swept by strong oceanic currents. It has large, thick, heavy scaling with iridescent white flecks, muscular paired fins, and an unusual tail with an additional central lobe. The pectoral fins of the coelacanth are highly mobile, and it uses these to maneuver into crevices to reach fishes and other prey. Populations are poorly documented and estimates are difficult, but susceptibility to capture by humans and a restricted habitat range have made the coelacanth an endangered species.

Latimeria menadoensis

Indonesian coelacanth

Location Pacific (Celebes Sea)

Length Up to 5¼ ft (1.6 m)
Weight 145–220 lb (65–98 kg)
Sex Male/Female
Status Vulnerable

The Indonesian coelacanth was discovered in the late 1990s, and as yet, relatively little is known about its behavior or ecology. However, since it is physically very similar to the coelacanth of southern Africa (see above), it almost certainly has a comparable way of life. Molecular analysis has shown that the 2 species probably diverged from each other between 4.7 and 6.3 million years ago. Since then, they have been kept apart by the geology of the seabed, and by currents that confine them to their own particular parts of the world. Like its African counterpart, the Indonesian coelacanth appears to be a solitary fish in open water, but it has also been found in groups in caves. So far, population estimates are not available, but it is likely to be endangered. The chief threat to this coelacanth's survival is fishing, which can have a severe effect on a species that is geographically highly isolated.

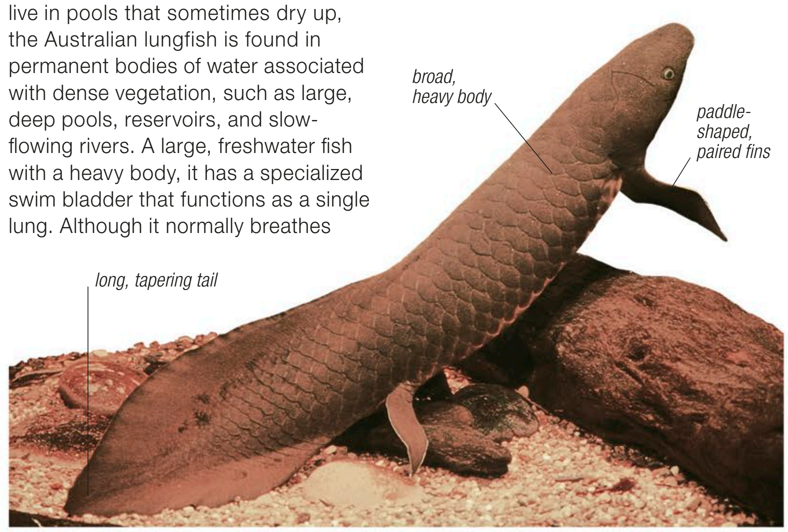
Neoceratodus forsteri

Australian lungfish

Location E. Australia

Length Up to 6 ft (1.8 m)
Weight Up to 99 lb (45 kg)
Sex Male/Female
Status Not evaluated

Unlike other lungfishes, which often live in pools that sometimes dry up, the Australian lungfish is found in permanent bodies of water associated with dense vegetation, such as large, deep pools, reservoirs, and slow-flowing rivers. A large, freshwater fish with a heavy body, it has a specialized swim bladder that functions as a single lung. Although it normally breathes through its gills, when oxygen levels in the water fall, it gulps air at the surface, breathing through its lung. Unlike African and South American lungfishes, this species has paddle-shaped, paired fins. Its dorsal and anal fins are continuous with its tail fin. The Australian lungfish becomes less active during dry conditions, and can survive for months if kept moist by a covering of damp leaves or mud. It feeds on frogs, crabs, insect larvae, mollusks, and small fishes, crushing its prey between its strong toothplates.



Protopterus annectens

West African lungfish

Location W. to C. Africa

Length 6–6½ ft (1.8–2 m)
Weight Up to 37 lb (17 kg)
Sex Male/Female
Status Least concern

The West African lungfish has long, threadlike, paired fins, and is the largest of the 4 lungfishes found in Africa. Like the other 3 species, it breathes through a pair of lungs. At the onset of the dry season, this fish burrows into mud, forming a mucus-filled cocoon. It lives in a variety of freshwater habitats, and is carnivorous, capturing its prey by stealth rather than by rapid pursuit.

FISHES